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Tissue-specific biosynthesis and spatial distribution of xanthones in *Garcinia oblongifolia* revealed by integrated metabolomic and full-length transcriptomic analyses

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Abstract

Background The medicinal plant *Garcinia oblongifolia* (Clusiaceae), widely used in the Lingnan area of southern China, exhibits notable anti-inflammatory, analgesic, antimyatrophy, antibacterial, and antitumor properties. Traditional applications, particularly the use of its bark, were extensive and largely attributed to bioactive xanthones. However, the tissue-specific distribution and biosynthesis of these compounds remained poorly understood.

Results Metabolite profiling identified 402 metabolites, including 215 that accumulated differentially across four tissues. Notably, 16 xanthones were predominantly localized in the bark and branches, with prenyl-substituted derivatives exhibiting the highest accumulation in branches. To elucidate xanthones biosynthesis, we integrated Illumina and single-molecule real-time (SMRT) sequencing, which yielded 70,401 full-length transcripts. Transcriptomic analysis revealed 8,842 differentially expressed transcripts, with xanthone-related genes showing bark-specific expression a pattern confirmed by qRT-PCR. We reconstructed a putative xanthone biosynthetic pathway involving 32 full-length transcripts encoding seven key enzymes. Phylogenetic and motif analyses revealed that four benzophenone synthase (BPS) homologs catalyze the conversion of benzoyl-CoA into 2,4,6-trihydroxybenzophenone. As a pivotal enzyme, BPS plays a central role in this biosynthetic pathway. Transcription factor analysis indicated that the bHLH family was closely associated with the regulation of BPS. Correlation and WGCNA analyses demonstrated a positive regulatory relationship between *PAL*, *C4H*, *CoA-hydratase* genes and their downstream metabolites. Notebly, *BPS*, *XS*, and *PT* genes were highly expressed in bark, while their corresponding metabolites accumulated in both bark and branches, suggesting inter-tissue translocation.

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Conclusions Our study analyzed xanthone distribution in *G. oblongifolia*, supporting its medicinal uses, and provided its first full-length transcriptome. We proposed a xanthone biosynthetic pathway, aiding in vitro synthesis and sustainable production, advancing pharmacological and biotechnological applications.

Keywords Full-length transcriptome, Garcinia oblongifolia, Metabolome, RNA-Seq, UHPLC-MS/MS, Xanthones biosynthesis

Introduction

Garcinia oblongifolia Champ. ex Benth, a species belonging to the genus *Garcinia* genus whitin the family Clusiaceae, is primarily distributed across tropical and subtropical regions, including Hainan Province, Guangdong Province, and Guangxi Zhuang Autonomous Region in China. Its wild resources of *G. oblongifolia* were especially abundant in the country of Wanning, Changjiang, Lingshui, and Ledong counties of Hainan Province. *G. oblongifolia* has been traditionally used by the Li and Zhuang ethnic groups in Lingnan area for its anti-inflammatory, analgesic, antimyatrophy, antibacterial, and antitumor properties [1, 2]. The bark of *G. oblongifolia* was employed in the treatment of ulcers, mild bleeding, stomatitis, and periodontitis. Furthermore, leaf extracts were demonstrated to be effective in treating wound decay [3]. This species also possessed considerable economic value: its fruits are edible and its seeds, bark, and wood serve as important industrial materials. Recent studies have shown that *G. oblongifolia* is a species rich in various bioactive compounds, including benzophenones, xanthones, flavonoids, triterpenes, and other compounds derived from bark of the tree showing promising applications in the fields of medicine, cosmetics, food, and health products.

Xanthones, important secondary metabolites in *G. oblongifolia*, were primarily distributed in the bark. These compounds exhibited a range of pharmacological effects, such as anti-tumor, antibacterial, antiviral, liver-protecting, blood-glucose-lowering, and the inhibition of xanthine oxidase [4]. In view of the fact that it is mainly distributed in the bark, which is not conducive to large-scale development and makes it difficult to meet the market demand. Therefore, elucidating the molecular mechanism underlying xanthone biosynthesis, identifying the biosynthetic genes, and clarifying its transcriptional regulatory patterns were considered essential for enabling rational resource utilization and applying biotechnology approaches to enhance the accumulation of these active compounds. As an important class of bioactive component, xanthones can be classified into six main groups: oxygenated xanthones, glycosylated xanthones, prenylated xanthones, xanthonolignoids, bis-xanthones, and miscellaneous xanthones [5]. Xanthones can primarily be isolated from approximately 20 species of higher plants, most of which belong to the families Clusiaceae, Hypericaceae, and Gentianaceae [6]. Currently, the complete biosynthetic pathway of xanthone has not yet been fully elucidated. Related studies have shown that xanthone biosynthesis in plants involves two key metabolic pathways: the phenylalanine and the shikimate pathways [7]. Critical enzymes mainly included phenylalanine ammonia-lyase (PAL), cinnamate: CoA ligase (CNL), cinnamoyl-CoA hydratase/lyase (CHL), benzaldehyde dehydrogenase (BD), benzoate-CoA ligase (BZL), 3-hydroxybenzoate-CoA ligase (3BZL), benzophenone synthase (BPS), benzophenone 3'-hydroxylase (B3'H), xanthone synthase (XS), various prenyltransferases (PT), and glycosyltransferases (GT) [7]. Among these, benzophenone synthase (BPS) played a central role by catalyzing the formation of 2,3',4,6-tetrahydroxybenzophenone, a key precursor in xanthone biosynthesis that connects both metabolic routes [8].

To date, multi-omics analysis has been widely used to investigate in the molecular mechanisms underlying the biosynthesis of active ingredients [9–12]. The integration of transcriptome and metabolome data has enabled the identification of genes involved in complex biological process with high sensitivity and accuracy [13–15]. In most transcriptome studies, short-read sequencing technologies such as Illumina transcriptome sequencing, have been commonly used. However, these technologies often pose challenges to the accuracy and integrity of transcript assembly [16]. Third-generation sequencing technologies enable the sequencing of full-length transcripts, but with lower sequencing accuracy [17]. Therefore, combining Illumina sequencing with PacBio full-length transcriptome sequencing can compensate for the shortcomings of each sequencing technology.Integrated metabolomic and transcriptomic analysis could reveal the regulation mechanisms of xanthone biosynthesis in *G. oblongifolia*.

In this study, we applied an integrated multi-omics approach to elucidate the xanthone biosynthesis pathway and identify key genes involved in *G. oblongifolia*. To the best of our knowledge, this is the first study to acquire full-length transcripts of *G. oblongifolia*, significantly improving the accuracy and depth of transcriptomic data interpretation. Given the complete absence of genomic resources for *G. oblongifolia*, our work establishes a foundational framework for clarifying xanthone biosynthesis, thereby facilitating future metabolic engineering and sustainable utilization of this pharmacologically important plant [17, 18].

Materials and methods

Plant materials

The sample collection for the current study was conducted in July, 2023 at Dagan Village, Qunying County, Lingshui County, Hainan Province, China (18°56′N, 109°5′E). Fresh tissue samples without pests, including bark (B), fruits (F), leaves (L), and branches (S) were obtained from three trees (biological replicates) of similar age (8 years), height (5.0 ± 0.3 m), trunk diameter (20 ± 1 cm), with uniform growth vigor were selected as biological replicates. The chosen individuals showed no visible disease, insect damage, or mechanical injury. All sampled trees shared similar environmental conditions, including slope position and soil type, to minimize the effects of environmental variation on experimental results. The plants material was formally identified by Dr. Bo Liu from College of Life and Environmental Sciences (CLES), Minzu University of China (MUC). Plant specimen (Collectors: Zhang Yi, Ma Yinyin, Li Shiyu, Collection No.: MUC_HNLS_230727_Go_1) are deposited in the Herbarium of Minzu University of China (MUCH). Subsequently, the fresh samples were immediately frozen in liquid nitrogen to preserve their biological integrity, stored at −80 °C, and subjected to metabolomic and sequencing analysis.

Metabolomic profiling and UHPLC-MS/MS analysis

Fresh samples of bark (B1, B2, and B3), fruits (F1, F2, and F3), leaves (L1, L2, and L3), and branches (S1, S2, and S3) were freeze-dried, pulverized into powder, and accurately weighed (25 mg each). Each powdered sample was then extracted with 0.5 ml of a methanol: water (3:1, v/v) solution. The mixtures were homogenized at 35 Hz for 4 min and followed by sonication for 5 min in an ice-water bath. This homogenization-sonication cycle was repeated 3 times. Subsequently, the samples were incubated at −40℃ for 1 h and then centrifuged at 12,000 rpm for 15 min. The supernatant was filtered through a 0.22 μm microporous membrane and collected into a new glass vial. A pooled quality control (QC) sample, representative of the entire sample set, was prepared by combining equal volumes of supernatants from all samples. This QC was analyzed throughout the acquisition sequence to monitor instrumental stability and data quality [19–22].

The filtered extracts were analyzed using an Ultra-performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) system (Vanquish, Thermo Fisher Scientific) equipped with a UPLC HSS T3 column (2.1 mm × 100 mm, 1.8 μm) and coupled to an Orbitrap Exploris 120 mass spectrometer (Orbitrap MS, Thermo). The mobile phase consisted of 5 mmol/L ammonium acetate and 5 mmol/L acetic acid in water (A phase) and acetonitrile (B phase). The autosampler temperature was set at 4℃, with an injection volume of 2 μL. The Orbitrap Exploris 120 mass spectrometer was chosen for its capability to acquire MS/MS spectra in information-dependent acquisition (IDA) mode using the Xcalibur acquisition software. Under these conditions, the full scan MS spectrum was continuously evaluated. The ESI source parameters were as follows: sheath gas flow rate at 50 Arb, Aux gas flow rate at 15 Arb, capillary temperature at 320 ℃, full MS resolution at 60,000, MS/MS resolution at 15,000, collision energy at 10/30/60 in NCE mode, and spray voltage at 3.8 kV (positive) or −3.4 kV (negative), respectively. The raw data were converted to the mzXML format using ProteoWizard and processed with an in-house program, which was developed using R and based on XCMS, for peak detection, extraction, alignment, and integration.

Metabolite analysis

To explore and visualize the variations among different sample groups, we utilized unsupervised dimensionality reduction through Principal Component Analysis (PCA) and hierarchical clustering analyses using R-based packages. Additionally, Orthogonal Partial Least Squares-Discriminant Analysis (OPLS-DA) was applied to the entire dataset. Variable Importance in Projection (VIP) scores derived from OPLS-DA were used to identify metabolites showing significant differential accumulation.

To validate the annotations of the metabolites, the peaks obtained were cross-referenced against publicly accessible mass spectrometry databases, including the Human Metabolome Database (HMDB) at http://www.hmdb.ca [23], Massbank at http://www.massbank.jp [24], LipidMaps at http://www.lipidmaps.org [25], and mzCloud at https://www.mzcloud.org [26]. Differentially accumulated metabolites (DAMs) were defined as those with VIP scores ≥ 1 and p-value from student's t-test < 0.05. The identified DAMs were mapped to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database at https://www.genome.jp/kegg [27] to enrich significant pathways. Further metabolite annotation was performed using an in-house MS2 database (BiotreeDB) with a cutoff of 0.8. The cutoff score of 0.8 for MS/MS matching was selected to stringently control false positive identifications while maintaining reasonable sensitivity, a common empirical standard in untargeted metabolomics [28–30]. The study aimed to identify specific substances involved in the xanthone biosynthesis pathway.

RNA extraction, illumina and PacBio sequencings

Frozen tissues were carefully transferred to a pre-cooled mortar and finely powdered using liquid nitrogen. Subsequently, total RNA was extracted from bark (samples B1, B2, and B3), fruits (samples F1, F2, and F3), leaves (samples L1, L2, and L3), and branches (samples S1, S2, and S3) of three individual plants. For this process, three

biological replicates were established, resulting in a total of 12 samples. The RNAprep Pure Plant Plus Kit (Tiangen, Beijing, China) was utilized to extract total RNA according to the manufacturer's guidelines. The total RNA underwent treatment with RNase-free DNase I (Takara, Beijing, China) to eliminate any contamination from genomic DNA. To assess the concentrations and purity of the extracted RNA, the Nanodrop 2000 and Agilent 2100 system (Agilent Technologies Inc, CA, USA) were employed. Furthermore, the quality and quantity of the RNA were also evaluated through agarose gel electrophoresis and spectrophotometry (IMPLEN, CA, USA). RNA samples with an A260/A280 ratio ranging between 1.8 and 2.2 were selected for the library construction and sequencing.

For Illumina sequencing, twelve individual libraries were prepared using the NEBNext® Ultra™ RNA Library Prep Kit (Illumina Inc), corresponding to bark (B1, B2, and B3), fruits (F1, F2, and F3), leaves (L1, L2, and L3), and branches (S1, S2, and S3) samples. The process began with enriching the mRNA from each sample using Oligo (dT) beads, followed by removing rRNA using RiboZeroTM Magnetic Kit (Epicentre). The enriched mRNA was then fragmented into smaller pieces and reverse transcribed into cDNA using random primers. After purification, the cDNA fragments underwent end repair, adapters ligation, and poly(A) tailing. The ligation products were then purified, PCR amplified, and sequenced on the high-throughput Illumina HiSeq™ 6000 platform (Novogene, Beijing, China), generating a set of paired-end reads with an average length of 150 bp.

For PacBio sequencing, a single library was constructed from equal amounts of total RNA from all twelve tissues, including bark (B1, B2, and B3), fruits (F1, F2, and F3), leaves (L1, L2, and L3), and branches (S1, S2, and S3).mRNA was enriched from total RNA usingoligo-dT magnetic beads, and the cDNA synthesis was carried out using theClontech SMARTer PCR cDNA Synthesis Kit (Takara Bio, Mountain View, CA, USA) for the SMRTbell library construction. PCR productswere size-selected to construct the nonselected and >4 kb selected cDNA libraries according to the BluePippin selection system. Additional PCR was performed to obtain sufficient cDNA.After repairing DNA damage, cDNA templates were ligated with sequencing adapters, primer, and polymerase. The library was then qualified and sequenced on the PacBio Sequel platform.

Transcriptome analysis and annotation

The Illumina sequences of raw data in fastq format from 12 samples were initially processed using in-house Perl scripts. In this step, clean data were obtained by removing reads containing adapter, reads containing poly-N and low quality reads from raw data. At the same time, Q30, GC-content and sequence duplication level of the clean data were calculated. All the downstream analyses were based on clean data with high quality.

The raw subreads of PacBio sequencing underwent analysis through the IsoSeq3 pipeline, available at https://github.com/PacificBiosciences/IsoSeq. This pipeline encompassed three key stages: creating circular consensus sequence (CCS) subreads, categorizing full-length (FL) reads, and clustering of full-length non-chimeric (FLNC) reads. CCS v6.2.0 was used to produce polished CCS subreads with a minimum quality threshold of 0.9 (min-rq 0.9). The default criterion of a minimum of 3 FL subreads (n = 3) was applied for CCS generation within a zero-mode waveguide (ZMW). Based on the presence of 5' and 3' adapters and poly(A) tails, sequences were classified as full-length or non-full-length. Subsequently, Lima v2.1.0 and IsoSeq3 refine were utilized to eliminate the primers and poly (A) tails, respectively. The isoform-level clusteringalgorithm was employed to generate high-quality FL consensus sequences, which were required to exhibit post-correction accuracy exceeding 99%. To cluster similar sequences within the full-length non-chimeric sequences, the IsoSeq3 module in SMRT-Link software was utilized, resulting in the generation of consensus sequences (consensus isoforms) for each cluster. The obtained consensus sequences were calibrated with the non-full-length sequences to obtain high-quality sequences for subsequent analysis. Following redundancy correction, cluster database at High Identity with Tolerance (CD-HIT) was used to filter out redundant consensus reads, resulting in the final non-redundant isoforms for subsequent analyses. These non-redundant transcripts served as the reference transcriptome for downstream investigations.

Using Kallisto software, the reads from the fq file were directly compared to the three generations of transcripts and counted to quantify the expression levels of the transcripts. The number of fragments extracted from a transcript correlated with the amount of sequencing data (or Mapped Data), transcript length, and transcript expression level. To ensure that the number of fragments accurately reflected the transcript expression level, it was essential to measure the Mapped Reads in the sample. Both the number of fragments and transcript length were normalized. Gene expression levels were normalized to fragments per kilobase of exon per million fragments mapped (FPKM) using Kallisto. Significance in differential transcript expression was determined based on a false discovery rate (FDR) threshold of < 0.01 and a fold change ≥ 2. The differentially expressed transcripts (DETs) were annotated following a BLAST search in the non-redundant protein database related to plants.

All predicted genes were annotated using public databases, including the NCBI non-redundant protein

database (NR) [31], Cluster of Orthologous Groups of proteins (COG) [32], euKaryotic Orthologous Groups (KOG) [33], Swiss-Prot [34], Protein family (*Pfam*) [35], Evolutionary Genealogy Graph (eggNOG) [36], Translation of EMBL(TrEMBL) [37], Gene Ontology (GO) [38], and Kyoto Encyclopedia of Genes and Genomes (KEGG) [27]. In this study, we primarily utilized Pfam domain analysis and NR annotation information to identify key enzyme genes involved in the xanthone biosynthetic pathway. Preliminary screening of transcriptome data was performed based on annotation information, followed by nucleic acid and amino acid sequence alignment of the initially screened sequences (E-value ≤ 1e-10), which led to the identification of enzyme gene transcripts associated with the xanthone biosynthetic pathway. The selected transcripts were aligned against the NCBI database to eliminate redundancies and subsequently the transcripts of enzyme genes pertinent to the synthesis pathway were identified.

Phylogenetic constructed and co-expression network analysis

The BPS (benzophenone synthase) protein sequences and their corresponding annotation files from multiple species were retrieved from the NCBI Protein database (https://www.ncbi.nlm.nih.gov/protein) [39] and UniProt (https://www.uniprot.org/) [40]. The full-length BPS prot ein sequence of *G.oblongifolia* was aligned using MEGA software (version 11.0.13). Subsequently, a phylogenetic tree was constructed based on the aligned sequences using the neighbor-joining (NJ) algorithm in Clustal W, with bootstrap analysis (1,000 replicates) performed to evaluate the robustness of the tree topology. The phylogenetic tree was visualized and refined using the Chiplot online platform (https://www.chiplot.online/).

Furthermore, the conserved motifs of downstream gene families were identified using MEME (version 4.5.5.7) (https://meme-suite.org/meme/tools/meme) and further annotated with ESPript (https://espript.ibcp.fr/ESPript/ESPript/index.php). Predict the transcription factors (TFs) of *G.oblongifolia* using iTAK (v1.7a). The correlation between key enzyme genes involved in the xanthone biosynthesis pathway and transcription factor families was statistically analyzed using GraphPad Prism 10, and the resulting interaction network was visualized in Cytoscape (version 3.10.3).

Integrated analysis of transcriptome and metabolome

The study utilized the Pearson correlation algorithm to assess the correlation between gene expression and metabolite response intensity data. Differential expressed transcripts (DETs) and differential accumulated metabolites (DAMs) were selected to create a correlation heatmap and matrix. Pathway analysis was conducted to investigate DETs and DAMs associated with xanthone biosynthesis. Weighted Gene Co-expression Network Analysis (WGCNA) was performed on the *G. oblongifolia* transcriptome datasets to identify potential transcript modules linked to major xanthone biosynthesis. WGCNA constructed networks based on the absolute value of Pearson's correlation coefficient, which was raised to a power to create the adjacency matrix. Correlation analysis between module eigengenes and tissue sites was utilized to explore the biological relevance of each module. Modules were generated using default settings in WGCNA Shiny within TBtools v1.098.

Quantitative real time PCR verification analysis

The expression profiles of nine genes associated with the xanthone biosynthesis pathway were randomly selected to validate the accuracy of transcriptomic data using qRT-PCR. Primers were designed using Primer-BLAST available on the NCBI website. The total RNA was reverse transcribed using the FastKing RT kit following the manufacturer's protocol. The qRT-PCR analysis was conducted on an ABI QuantStudio 3 platform with three biological and three technical replicates for each sample. The qRT-PCR reaction system (20 μL) comprised 10 μL of Universal SYBR Green Fast qPCR Mix, 1 μL of cDNA, 0.4 μL each of forward and reverse primers, and 8.2 μL of sterile water. The qRT-PCR protocol involved an initial denaturation step for 3 min, followed by 40 cycles at 95 °C for 5 s, 60 °C for 30 s, and 72 °C for 12 s. The relative expression levels were determined using the $2^{-\Delta\Delta}$Ct method [41] and normalized to the actin gene [42, 43].

Results

Metabolomic analysis and screening of Xanthone substances

A total of 402 metabolites meeting the cutoff of 0.8 were identified in positive and negative ionization modes (Table S1), including 193 shikimates and phenylpropanoids, 49 terpenoids, 44 alkaloids, 42 fatty acids, 21 polyketides, 21 amino acids and peptides, 19 carbohydrates, and 13 other unknown metabolites. Shikimates and phenylpropanoids (48.010%) and terpenoids (12.189%) constituted a significant proportion of all metabolites (Fig. 1A). The heatmap of sample correlations (sample-to-sample clustering) indicated that metabolite accumulation values across 12 samples of *G. oblongifolia* were consistent among the three biological replicates (Fig. 1B). These findings demonstrated the good reproducibility of different tissue parts of *G. oblongifolia* and the reliability of the experimental outcomes. A strong correlation was observed between the branches and bark samples, possibly linked to the types and quantities of metabolites present. In the PCA score plot (Fig. 1C), the high degree of clustering observed

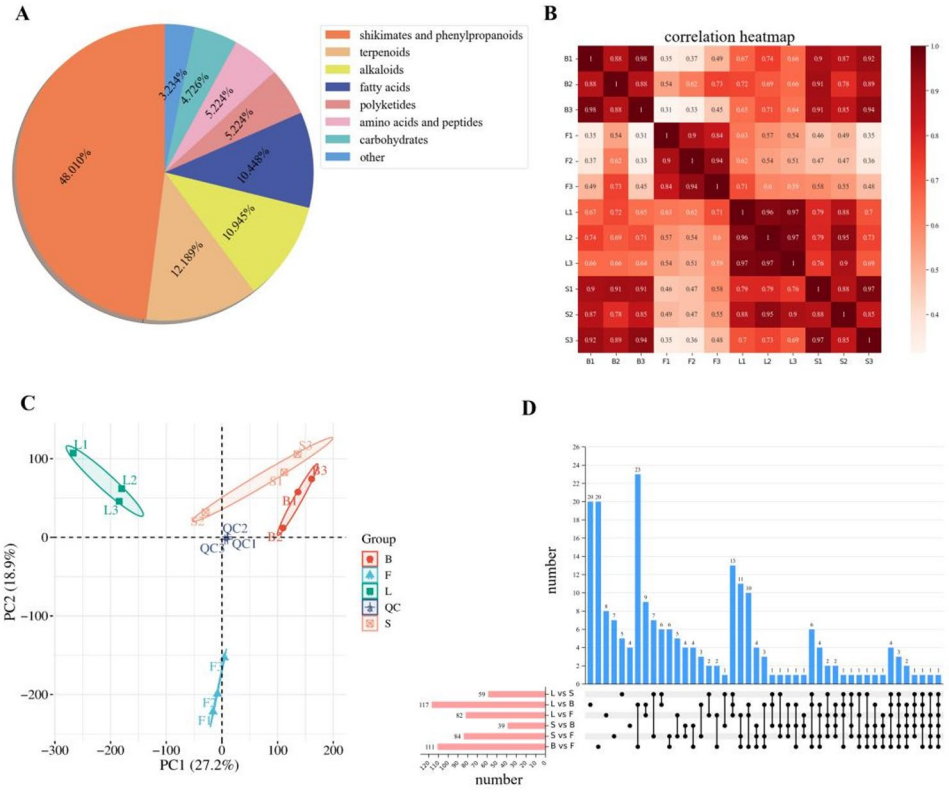


Fig. 1 Metabolite accumulation of different tissues in *G. oblongifolia*. Note: Bark (B1, B2, and B3), fruits (F1, F2, and F3), leaves (L1, L2, and L3), branches (S1, S2, and S3) and quaility control (QC1, QC2 and QC3) **A** Classification of the metabolite of *G. oblongifolia* samples **B** Correlation heatmap of gene expression profiles across different tissue types in *G. oblongifolia* **C** PCA scores plot of different parts of *G. oblongifolia* **D** Upset plot illustrating the unique and shared metabolites across different tissue types in *G. oblongifolia*

among the QC samples underscored the robust stability of the analytical method and the high quality of the data obtained. *G. oblongifolia* samples showed a significant trend of metabolic separation between different samples, indicating significant differences among various parts of *G. oblongifolia.* However, it was important to acknowledge that the performance of biological replicates in the branches section was suboptimal, likely due to variations in branches growth years and differences in plant part selection for sampling. Additionally, a relatively high correlation (Fig. 1B) and little difference (Fig. 1D) was observed between branches and bark, offering valuable insights for the collection of plant materials.

To examine metabolite variation across different parts of *G. oblongfolia*, K-means clustering was utilized to group metabolites based on their expression profiles, resulting in the identification of nine distinct clusters (Fig.S1). Cluster 3 and 4 predominantly contained shikimates and phenylpropanoids, with higher accumulation levels observed in branches and bark compared to leaves and fruits. Concurrently, clusters 2, 5, and 7 also contained a certain proportion of shikimates and phenylpropanoids, predominantly flavonoids. This finding substantiated the presence of elevated expression levels of certain specific shikimate and phenylpropanoid compounds within the leaves tissue.

To explore metabolic differences in various parts of *G. oblongifolia*, a screening of 402 metabolites was conducted based on VIP value, P value, and FC value criteria. Metabolites with a VIP value ≥ 1, P value < 0.05, and FC value ≥ 2 or ≤ 0.5 were considered for analysis. The results were visualized using an Upset plot (Fig. 1D) and Volcano plots (Fig.S2). The number and regulation direction of differential metabolites for each comparison were summarized in Fig. 3. Furthermore, unique and shared metabolites were identified, with the highest number of shared DAMs observed between L vs. B and B vs. F, along with a notable number of unique DAMs in these two groups compared to others (Fig. 1D).

The differential metabolites identified in each comparison group were mapped to the KEGG database, revealing enrichment in 45, 45, 28, 15, 12, and 30 pathways in B vs. F, L vs. B, L vs. F, L vs. S, S vs. B, and S vs. F, respectively. The prominent pathways were visualized in Bubble plots (Fig.S3). Significant enrichment pathways (p-value < 0.05) were observed in different comparison group, such as biosynthesis of various plant secondary metabolites and phenylpropanoid biosynthesis between L vs. B,

beta-Alanine metabolites and phenylalanine metabolites between L vs. F, vitamin B6 metabolites and ubiquinone and other terpenoid-quinone biosynthesis between S vs. F, glyoxylate and dicarboxylate metabolites and D-Amino acid metabolites between B vs. F, carotenoid biosynthesis and biosynthesis of various plant secondary metabolites between L vs. S, and biosynthesis of various alkaloids and aminoacyl-tRNA biosynthesis between S vs. B.

Xanthones, a class of compounds derived from shikimates and phenylpropanoids include six main groups. Prenylated xanthones, in particular, exhibit significant biological and pharmacological activities, with *Garcinia* spp. being a notable natural source. In this study, 16 xanthones were identified in *G. oblongifolia* (Fig. S4), with 10 of them being prenylated xanthones. A cluster heatmap analysis of these xanthones revealed their predominant distribution in bark, branches, and leaves, with lower levels in fruits, consistent with previous research findings [44] (Fig.S5). Notably, the accumulation level of 8 prenylated xanthones, such as ugaxanthone, cratoxyarborenone e, morusignin b, macluroxanthone, mangostin, dulxanthone f, garcinone b and gamma-mangostin, were found to be higher in bark and branches than other tissues, while the other two prenylated xanthones enriched in fruits. Two oxygenated xanthones, including lichexanthone and norathyriol, were significantly enriched in the branches. In contrast, the other 4 glycosylated xanthones, particularly mangiferin and tripteroside, were significantly enriched in leaves (Fig.S5).

Quality control analysis of transcriptome data

Both the NGS Illumina and SMRT PacBio sequencing platforms were used to identify the transcripts of enzyme genes pertinent to the xanthone biosynthesis pathway. First, 12 mRNA samples from 4 different tissues (bark, fruits, leaves and branches) were subjected to 2 × 150 paired-end sequencing using the HiSeq 2500 platform, producing 247,443,505 reads (Table S2).The adapters and primer sequences were cut off in reads, and the low-quality value data were filtered, resulting in the acquisition of high-quality reads.A total of 74.11 Gb of clean bases were generated from Illumina sequencing platform after data cleaning and quality assessment, and each sample yielded more than 5.77 Gb of clean data. Q30 value of all 12 samples were over 97.18%, indicating no more than 2.82% of bases with a sequencing error rate below 1‰. The GC content of the 12 samples clean data ranged from 43.25% to 44.87% (Table S2). These results confirmed the reliability and suitability of the sequencing data for further analysis. The full-length cDNAs from twelve pooled poly (A) RNA samples were normalized and subjected to a SMRT sequencing using the PacBio platform and 22.68 Gb of data were generated. A total of 280,945 CCS reads were generated after extracting the CCS sequence from the original sequence according to the condition that full passes ≥ 3 with the sequence accuracy greater than 0.9 and correcting the CCS sequences. Subsequently, FLNC reads were identified by the presence of 5′-primers, 3′-primers, and poly (A) tails, resulting in the detection of 226,968 FLNC reads (80.79% of CCS) with an average length of 1,485 bp.

To address the high error rate of PacBio reads and enhance accuracy, IsoSeq3 modules in the SMRT Link software were utilized to cluster similar full-length nonchimeric sequences into consensus isoforms. This process resulted in the generation of 109,124 full-length consensus isoforms, consisting of 109,098 high-quality (HQ) and 26 low-quality (LQ) transcripts. After removing redundant sequences using the CD-HIT program, a total of 70,401 nonredundant transcript isoforms were obtained.

Table 1 Number of transcripts annotated in different database

Annotated databases	Transcripts Number	Transcripts Percent (%)
NR	52,665	74.81%
COG	15,137	21.50%
KOG	33,182	47.13%
Swiss-Port	35,280	50.11%
Pfam	36,773	52.23%
eggNOG	45,399	64.49%
TrEMBL	52,115	74.03%
GO	43,411	61.66%
KEGG	38,062	54.06%
ALL	53,358	75.79%

Functional annotation of the *G.oblongifolia* transcriptome data

To comprehensively annotate the *G. oblongifolia* transcriptome, 53,358 (75.79%) out of 70,401 full-length transcripts were annotated by at least one of the following 7 databases, including nucleotide and protein databases (NR, COG, KOG, Swiss-Prot, Pfam, eggNOG, TrEMBL, GO, and KEGG) (Table 1). Additionally, the unannotated transcripts may represent novel genes in *G. oblongifolia*. BlastX similarity analysis against the NR database revealed that the *G. oblongifolia* full-length transcripts exhibited similarity to several plant species (Fig. 2A). Among them, 7,818 (14.84%) transcripts showed significant homology with *Hevea brasiliensis*, while 5,515 (10.47%) and 4,129 (7.84%) transcripts displayed high similarity with sequences of *Jatropha curcas* and *Manihot esculenta*, respectively.

The COG analysis revealed that 15,137 transcripts were grouped into 25 functional clusters. The five largest categories identified were: 'translation, ribosomal structure and biogenesis' (1,778 transcripts, 11.9%),

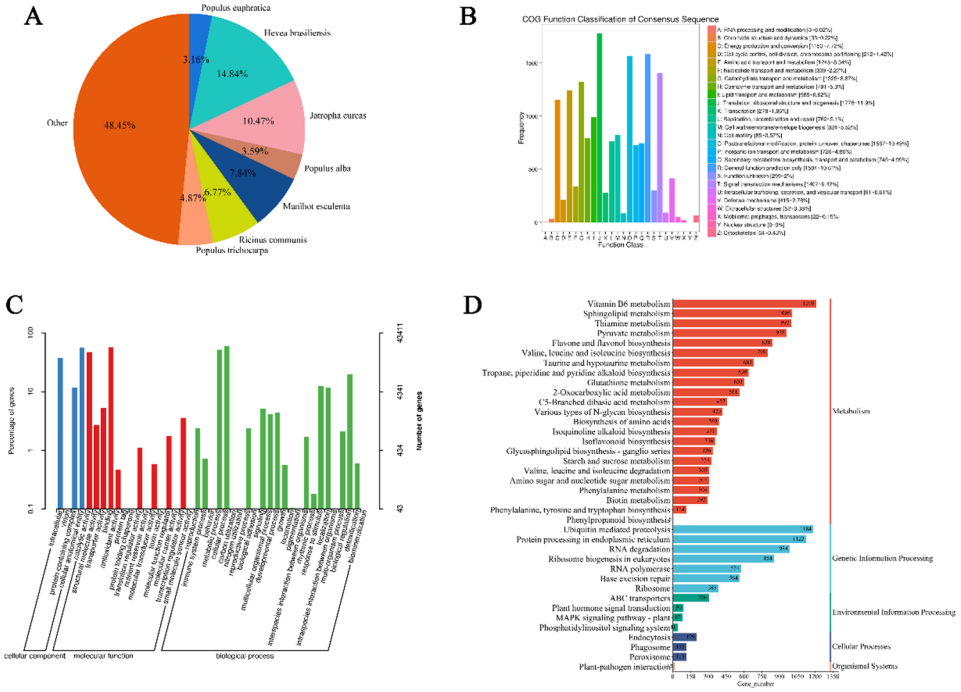


Fig. 2 Function annotation and classification of full-length transcripts in *G. oblongifolia* **A** NR classifcation analysis of transcripts **B** COG enrichment analysis of transcripts **C** GO classification analysis of transcripts. The annotated transcripts were distributed in biological process, cellular component and molecular function **D** KEGG enrichment analysis of transcripts

'general function prediction only' (1,584 transcripts, 10.61%), 'posttranslational modification, protein turnover, chaperone' (1,567 transcripts, 10.49%), 'signal transduction mechanisms' (1,407 transcripts, 9.42%), and 'carbohydrate transport and metabolism' (1,325 transcripts, 8.87%) (Fig. 2B). Subsequently, GO annotation of the transcripts categorized them into three major groups: biological process, molecular function, and cellular component. Within the group of biological processes, there were 24 subclasses including metabolic process (22,166 transcripts), biological regulation (8,530 transcripts), and developmental process (1,908 transcripts) and others. Molecular function which had 15 subclasses including catalytic activity (20,504 transcripts), transporter activity (2,285 transcripts), transcription regulator activity (1,563 transcripts), and others. The cellular component group comprised 4 subclasses: cell anatomical entity (24,145 transcripts), intracellular component (16,527 transcripts), protein-containing complex (5,129 transcripts), and virion (2 transcripts) (Fig. 2C).

In the context of KEGG analysis, it proves beneficial for identifying functional genes, comprehending gene functions, and exploring gene interactions within biosynthetic pathways. A total of 38,062 transcripts from *G. oblongifolia* were annotated in the KEGG database. These transcripts were clustered into 138 pathways belonging to five major categories, metabolism (107 pathways), genetic information processing (21 pathways), cellular processes (4 pathways), environmental information processing (4 pathways), and organism systems (2 pathways). The most extensive pathways were metabolism, especially the primary metabolism, encompassing 1,208, 1,006 and 997 transcripts enriched in vitamin B6 metabolism, sphingolipid metabolism and thiamine metabolism, respectively. Furthermore, some transcripts were linked to significant metabolism pathways like flavone and flavonol biosynthesis (838 transcripts), phenylalanine metabolism (304 transcripts) and phenylpropanoid biosynthesis (5 transcripts) (Fig. 2D). Given the strong connection between phenylalanine and xanthone biosynthesis, our focus was directed towards these to discover the potential pathway.

Analysis of differentially expressed genes and differentially accumulated metabolites in different tissues

To investigate the variation in transcript abundance and expression patterns among different tissues of *G. oblongifolia*. The FPKM distribution across all samples was illustrated in Fig.S6 Six comparison groups were established to examin transcriptomic differences in different tissue sites of *G. oblongifolia*: S vs. B, L vs. S, S vs. F, L vs. F, L vs. B, and B vs. F (Fig. 3 and Table S3). Overall, 261 transcripts (30 up-regulated and 231 down-regulated), 750 transcripts (546 up-regulated and 204 down-regulated), 774 transcripts (440 up-regulated and 334 down-regulated), 1,949 transcripts (1,283 up-regulated and 666 down-regulated), 2,351 transcripts (1,155 up-regulated

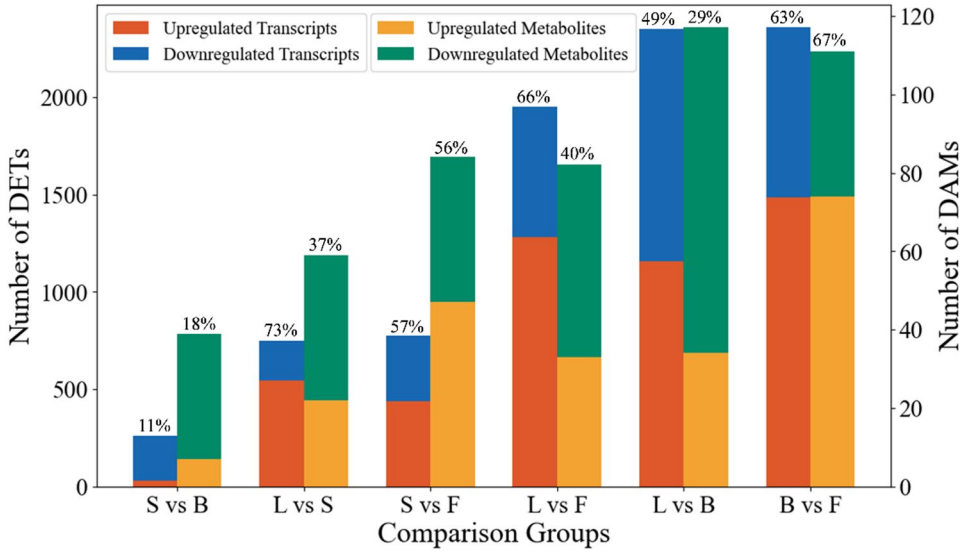


Fig. 3 The number of differentially expressed transcripts and differentially accumulated metabolites. The left and right ordinates represented the number of differentially expressed transcripts and differentially accumulated metabolites in each compared tissue types. The abscissa represented six comparison groups. S vs. B: branches vs. bark, L vs. S: leaves vs. branches, S vs. F: branches vs. fruits, L vs. F: leaves vs. fruits, L vs. B: leaves vs. bark, and B vs. F: bark vs. fruits. The percentage at the top of the bar indicated the proportion of the up-regulated transcripts in DETs or up-regulated metabolites in DAMs of each comparasion group

and 1,196 down-regulated), and 2,357 transcripts (1,487 up-regulated and 870 down-regulated) transcripts were differentially accumulated in S vs. B, L vs. S, S vs. F, L vs. F, L vs. B and B vs. F (Fig. 3). A pronounced concordance was observed between the transcriptomic and metabolomic profiles, wherein comparison groups exhibiting a higher number of differentially expressed genes in the transcriptome correspondingly displayed a greater abundance of differential metabolites in the metabolome. The percentage of up-regulated and down-regulated DETs were also consistent with that of DAMs in the comparison groups of S vs. B, S vs. F, and B vs. F (Fig. 3). In the comparison group L vs. B, the percentage of up-regulated DETs was 20% more than that of up-regulated DAMs. Furthermore, the proportion of up-regulation and down-regulation of DETs and DAMs was reversed in the Lvs. S and L vs. F comparison groups, which indicated that the genes up-regulated in leaves were not highly correlated with the production of metabolites, such as the genes in photosynthetic system.

A venn diagram of the differentially expressed transcripts in each comparison group was created to analyze the unique and common differentially expressed genes in each comparison group. The results revealed that the fruits, leaves, bark, and branches contained 302, 239, 105, and 9 specially expressed transcripts, respectively (Fig.S7). A total of 8,442 genes were differentially expressed across all comparison groups, suggesting a potentially crucial role in the metabolism of different tissues in *G. oblongifolia*. GO and KEGG enrichment analyses were conducted to further investigate the identified transcripts (Fig.S8 and Fig.S9). The results revealed that a broad range of metabolic pathways presented in the four different tissues of *G. oblongifolia*. In each comparison group, the differential genes were mainly enriched in carbon metabolism, phenylpropanoid biosynthesis, plant hormone signal transduction, plant-pathogen interaction, and phenylalanine metabolism.

Identification of full length transcripts putatively involved in Xanthone biosynthesis pathway

Xanthones are a diverse group of secondary metabolites with various pharmacological properties, including antitumor, antidiabetic, and antimicrobial effects. These compounds are synthesized in plants via the shikimate pathway and phenylalanine dependent pathway. In Hypericaceae, the biosynthesis of xanthone primarily occurs utilizing L-phenylalanine as a substrate, while in Gentianaceae, shikimate is converted to 3-hydroxybenzoic acid for further reactions (Fig. 4). This study identified 32 full-length transcripts encoding 7 enzymes involved in xanthone biosynthesis, including PAL (5 transcripts), CoA hydratase (9 transcripts), C4H (1 transcripts), 4CL (6 transcripts), BPS (5 transcripts), XS (5 transcripts), and PT (1 transcripts) (Table 2). The presence of multiple transcripts assigned to the same enzyme suggested that they might represent different isoforms of a single gene. The enzyme gene sequences identified in this study shared certain similarities with those from other known species, indicating potential conservation in xanthone biosynthesis among different species.

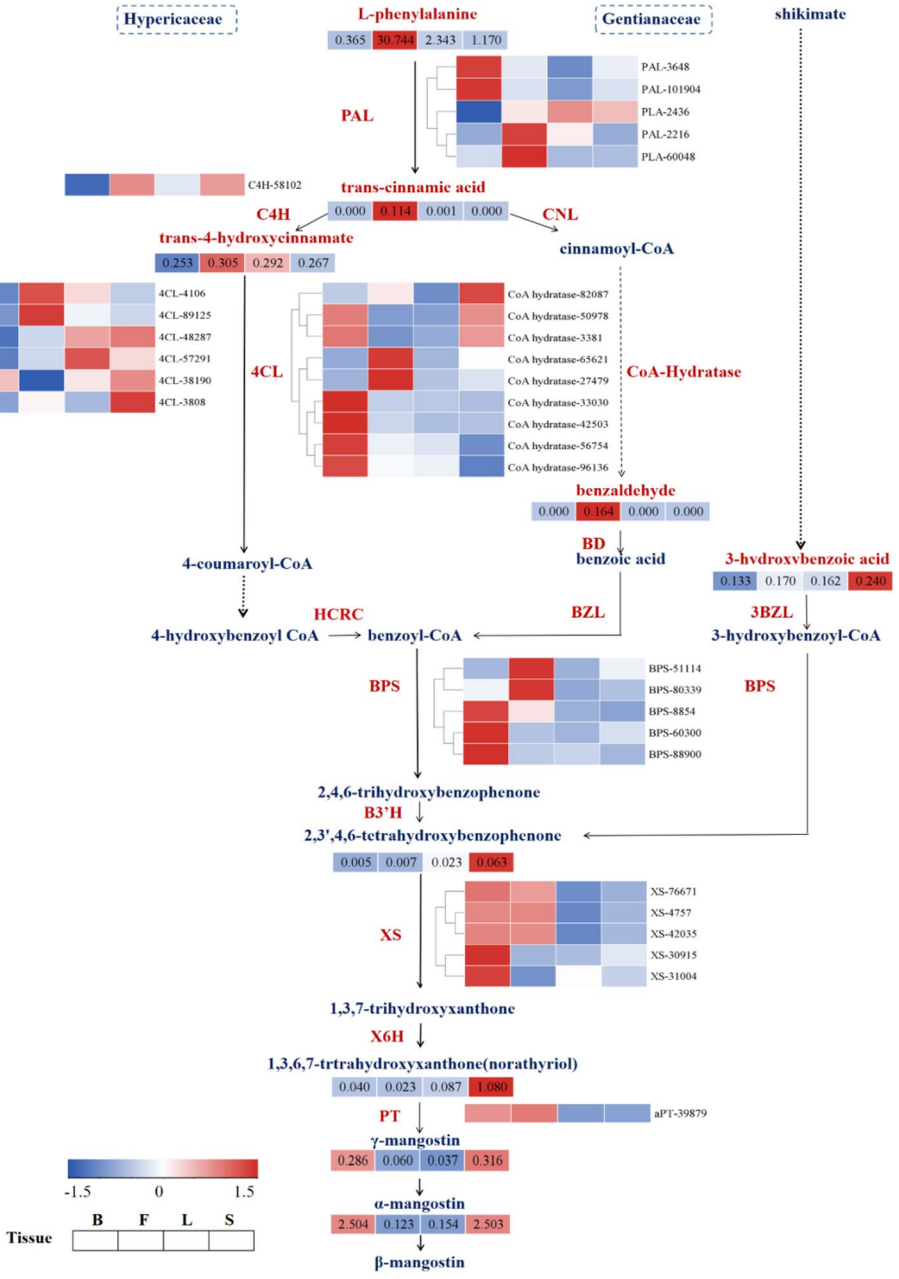


Fig. 4 Putative biosynthetic pathways of xanthone in *G. oblongifolia*. The tissue expression of key enzyme genes and metabolites in bark, fruits, leaves and branches of the synthesis pathway and the proposed biosynthetic pathways of several xanthone derivatives derived from 1,3,7-trihydroxyxanthone precursor, mainly involved the relevant xanthone obtained in this experiment. Full names of enzymes represented by their abbreviated names were showed. PAL: phenylalanine ammonia lyase, CNL: cinnamate-CoA ligase, C4H: cinnamate 4-hydroxylase, 4CL: 4-coumarate-CoA ligase, BD: benzaldehyde dehydrogenase, BZL: benzoate-CoA ligase, 3BZL: 3-hydroxybenzoate-CoA ligase, HCRC:4-hydroxybenzoyl-CoA reductase subunit gamma, BPS: benzophenone synthase, B3'H: benzophenone 3'-hydroxylase, XS: xanthone synthase, X6H: xanthone-6-hydroxylases, PT: prenyltransferase. The dotted line indicates that the enzymes involved in this pathway are unknown, and there may also be a multi-step reaction process. The expression levels deduced from the FPKM of each transcript that encodes the relevant enzyme, and relative accumulation levels of metabolites were shown as heat map, whereas bark, fruits, leaves and branches were separately analyzed. The expression level was calculated based on the Log2 Fold change > 1. Red and blue represented high and low expression level, respectively

Interestingly, our findings suggested the existence of an alternative pathway in *G. oblongifolia* that differed from that in Hypericaceae, because we did not detect the presence of CNL enzyme in the transcriptome sequence. L-phenylalanine, which can be converted from shikimate, is firstly converted to trans-cinnamic acid by the PAL enzyme, and is converted to trans-4-hydroxycinnamate by C4H enzyme, and then under the catalysis of 4CL,

Table 2 List of putative transcriptsinvolved in Xanthone biosynthesis pathway

Name	Description	The number of transcripts
PAL	phenylalanine ammonia lyase	5
CoA hydratase/lyase	CoA hydratase/lyase	9
C4H	cinnamate-4-hydroxylase	1
4CL	4-coumarate–CoA ligase	6
BPS	benzophenone synthase	5
XS	xanthone synthase	5
aPT	prenyltransferase	1

4-coumaroyl-CoA is generated. After a series of reactions similar to the two steps in the β-oxidation pathway of fatty acids, 4-hydroxybenzoyl-CoA is generated, and then under the catalyzation of 4-hydroxybenzoyl-CoA reductase subunit gamma (HCRC), benzoyl-CoA is generated. Benzoyl-CoA is an important intermediate metabolites of xanthone. After the reactions of BPS, benzoyl-CoA could transfer to 2,4,6-trihydroxybenzophenone, and then could produce 2,3',4,6-tetrahydroxybenzophenone catalyzed by B3'H (Fig. 4). The compound 2,3',4,6-tetrahydroxybenzophenone is the direct precursor of xanthone, and it is catalyzed by xanthone synthase to produce 1,3,5-trihydroxyxanthone (1,3,5-triHX) or 1,3,7-trihydroxyxanthone (1,3,7-triHX), which can be further converted into various xanthone compounds. To further investigate the regulation of key enzyme genes involved in the xanthone biosynthesis pathway, the intermediate metabolites within this pathway were examined and 5 intermediate metabolites, including L-phenylalanine, trans-cinnamic acid, benzaldehyde, trans-4-hydroxycinnamic acid, and 2,3',4,6-tetrahydroxybenzophenone were presented in a heat map alongside their corresponding regulatory enzymes genes to support this metabolic pathway (Fig. 4). In addition, we also detected 3-hydroxybenzoic acid, which could be in the shikimate pathways and could also be catalyzed by BPS and directly generate 2,3',4,6-tetrahydroxybenzophenone, which indicated that the biosynthesis of xanthone in *G. oblongifolia* involved both phenylalanine and shikimate pathways. According to the annotation information of them, the transcripts were renamed with the gene name plus transcript sequence number (Table S4).

Based on the xanthone biosynthetic pathway diagram and an analysis of the key enzyme genes and intermediate metabolites, we found that the PAL, C4H and 4CL enzyme gene were highly expressed in fruits tissues. Concurrently, L-phenylalanine, trans-cinnamic acid and trans-4-hydroxycinnamate were also highly accumulated in fruits tissues. The accumulation level of benzaldehyde aligned closely with the tissue expression level of the related CoA-Hydratase enzyme genes. The BPS enzyme gene was highly expressed in bark and branches. We detected the content of 2,3',4,6-tetrahydroxybenzophenone and found that it mainly accumulated in the branches. For the xanthone synthase detected in this study all belonged to 1,3,7-trihydroxyxanthone synthase. Furthermore, like BPS genes expression patterns, most xanthone synthase genes were predominantly expressed in bark and fruits, and it was speculated that this process involved material migration from bark and fruits to branches (Fig. 4).

The 1,3,5-trihydroxyxanthone and 1,3,7-trihydroxyxanthone serve as substrates for xanthone compounds such as mangostin and mangiferin synthesis [7, 45]. Through structural analysis of 16 xanthone compounds, it was determined that only a few screened xanthones were oxygenated xanthones and glycosylated xanthones, and the majority of the screened xanthones were prenylated xanthones (Fig.S5 and Fig.S10), which could be classified based on the the number of isopentenyl substitutions, such as mono-isopentenyl-substituted xanthones and di-isopentenyl-substituted xanthones. Therefore, prenyltransferases (PTs) were focused in this study. The prenylation of small aromatic molecules is a crucial step in the biosynthesis of various natural products. This process involves linking isoprenoid precursors, obtained from either the mevalonate or methylerythritol phosphate pathway, with aromatic precursors, primarily derived from the polyketide or shikimate pathway. Such interactions have led to the development of a wide array of secondary metabolites in bacteria, fungi, and plants [46]. Consequently, PT genes were expected to play essential roles in the synthesis of numerous xanthones in *G. oblongifolia*. Despite this, there was little study on the PTs in *G. oblongifolia*. In this study, we identified 1 prenyltransferases (PTs) genes from the transcriptomes using Pfam annotation and the BLAST algorithm. Among these, most genes belonged to the PT-UbiA family, and the aromatic prenyltransferase genes reported so far from plants all belonged to the UbiA superfamily [47]. Many well-characterized members of the UbiA prenyltransferase family are aromatic prenyltransferases that play a crucial role in the biosynthesis of compounds [48]. The annotation of these 1 prenyltransferases revealed the presence of only one aromatic prenyltransferase (Table S5).

Protein sequence alignment, phylogenetic, and transcription factor regulatory analysis of the *G.oblongifolia* BPS family

To investigate the potential biological functions of benzophenone synthase (BPS), we constructed a phylogenetic tree of BPS proteins and a TF co-expression network. The phylogenetic analysis revealed that most BPS members from *Garcinia oblongifolia* and *Garcinia mangostana* clustered together (see Supplementary Table S6

for detailed sequence information). The 22 BPS protein sequences clustered into three major clades (Fig. 5A; Supplementary Table S6). The 22 BPS protein sequences all belong to the legume branch among rosid plants [49]. Most notably, five BPS *Garcinia oblongifolia* and four *Garcinia mangostana* BPS protein formed a subclade with four *Hypericum androsaemum* sequences, all belonging to the order Malpighiales. Within this group, *Garcinia oblongifolia* BPS pairs (*Go*-BPS-8854/80339 and *Go*-BPS-60300/88900) exhibited strong bootstrap supported (94.2% – 100%) and high sequence homology with the *G. mangostana* (*Gm*-L7NCQ3.1 and *Gm*-AEI27291.1). Meanwhile, four *Malus domestica* BPS sequences (Rosaceae, Rosales) grouped into a seperate clade, though Rosales and Malpighiales both belong to the Fabids lineage. Intriguingly, two *H. androsaemum* BPS proteins (Ha-Q9FUB7.1 and Ha-AAG30295.1) clustered with *M. domestica* sequences (Md-5UC5|B and Md-5UC5|A), suggesting a close evolutionary relationship between Hypericaceae (Malpighiales) and Rosaceae (Rosales). Additionally, one *G oblongifolia* BPS specifically grouped with *H. androsaemum*, while two *Malus domestica* BPS showed high conservation with *Hypericum*, further supporting shared ancestry. These findings highlight the genetic affinities among Clusiaceae, Hypericaceae, and Rosaceae.

To further elucidate phylogenetic relationships, we performed sequence alignment and conserved domain analysis on four *G. oblongifolia* BPS proteins and two functional characterized *G. mangostana* BPS protein sequences (Gm-AEI27291.1 and Gm-L7NCQ3.1) [8]. Transcriptome-based alignment revealed >80% amino acid sequence similarity among these six proteins (Fig. 5B). Conserved motif analysis showed identical architectural patterns, with all sequences containing seven conserved motifs (Motifs 1–7) arranged in the same order (Fig. 5C). Particularly, Motif 1 (highlighted in red) displayed the highest conservation (E-value of $2.7e^{-203}$), with all seven motifs being universally present across all six BPS protein sequences (Fig. 5C).

In this study, a total of 1,709 transcription factors (TFs) from *G. oblongifolia* encompassing major families such as WRKY, C2H2, NAC, AP2/ERF-ERF, MYB, and bHLH (Fig. S11). Given the critical role of TFs like bHLH in

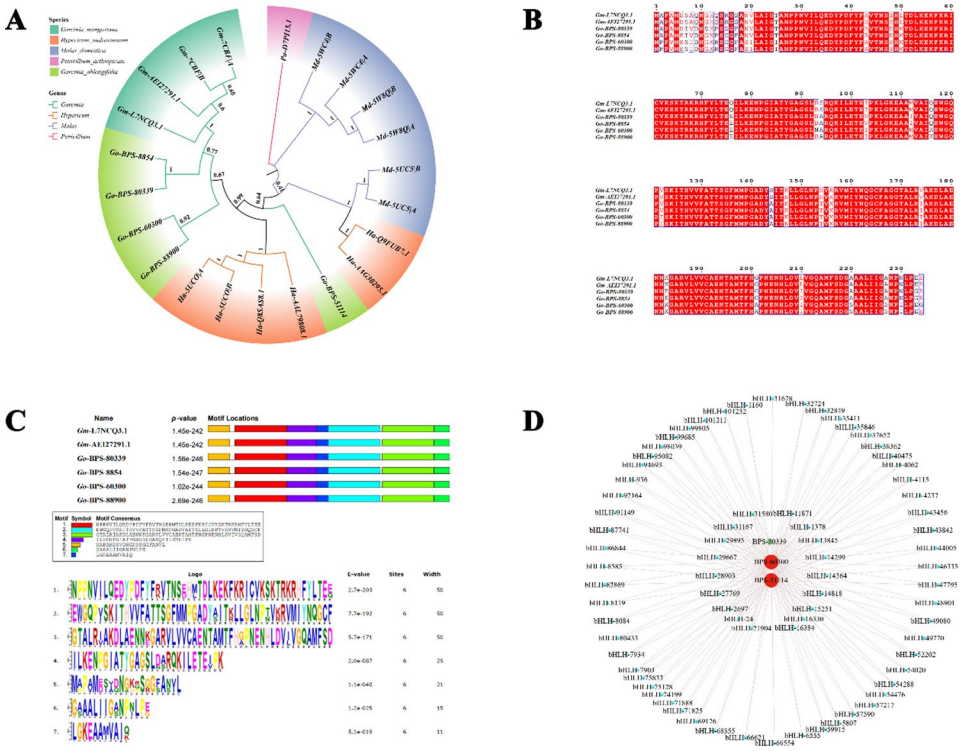


Fig. 5 Phylogenetic tree construction, conserved motif analysis, and transcription factor regulatory network analysis of BPS genes in *G. oblongifolia* **A** Phylogenetic analysis of BPS gene family in higher plants. *Go:Garcinia oblongifolia*; *Gm:Garcinia mangostana*; *Ha:Hypericum androsaemum*; *Md:Malus domestica*; *Pa:Penicillium aethiopicum* **B** Sequence alignment of BPS transcripts between *G. ablongifolia* and *G. mangostana*. The area with a red background represents the conserved part of the sequence. The white letters indicate the bases or amino acids at the corresponding positions, and there are differences in the base sequences within this area. The numerical labels above are used to indicate the positions of the sequences, facilitating the observation of the similarities and differences of different gene sequences at the same positions **C** Motif analysis results of *G. ablongifolia* and *G. mangostana* **D** Co-expression network between bHLH and xanthone biosynthesis genes in *G. oblongifolia*. The connections between nodes indicate that genes have co-expression relationships, and the size of nodes is proportional to the correlation coefficient

regulating secondary metabolites biosynthesis [50, 51] and the bioactivity of xanthones in *G. oblongifolia*,we analyzed the correlation between screened BPS transcripts and 86 bHLH TFs, revealing 75 bHLH TFs were associated with three key BPS transcripts (BPS-80339, BPS-51114, and BPS-60300), including 18 bHLH TFs with particularly strong regulatory links (Fig. 5D). The 18 identified bHLH TFs were classified into five functional categories (Table S7). The first category comprised TFs involved in flavonoid biosynthesis regulation, including homologs of bHLH35 in *Cucumis melo*, bHLH93 in Jujube, bHLH74 in *Actinidia chinensis* and SPATULA in tomato [52–56]. Three categories were associated with defense signaling pathways: (1) jasmonic acid-mediated defense response, represented by bHLH13 in *Arabidopsis thaliana* [57], (2) salicylic acid-dependent defense signaling, represented by TF ILR3 in *Arabidopsis thaliana* [58, 59], and (3) defense against fruit flies, represented by bHLH162 in *Citrus sinensis* [60]. The fifth category included TFs regulating pollen and anther development, exemplified by bHLH089 in *Arabidopsis thaliana* [61].Tissue-specific expression analysis revealed distinct expression patterns among these bHLH TFs. BPS-80,339 and BPS-51,114 showed fruit-predominant expression, while BPS-60,300 was bark-enriched. Among the 18 bHLH TFs, four were fruit-biased (bHLH-14818, bHLH-16384, bHLH-21904, bHLH-13845) (Fig. S12), whereas bHLH-28,903 demonstrated particularly high expression levels in bark and branch tissues.

Correlation analysis and weighted gene Co-expression network analysis reveal genes highly correlated with Xanthone biosynthesis

By comparing the DETs and DAMs in each group, it was observed that many DETs and DAMs were enriched in the same KEGG pathways (Fig.S13), with the pathway descriptions listed in Table S8. Based on the six comparison groups, the main enriched pathways for differential genes and differential metabolites included the biosynthesis of cofactors, biosynthesis of amino acids, plant hormone signal transduction, phenylpropanoid biosynthesis, ubiquinone and other terpenoid-quinone biosynthesis, phenylalanine metabolism and biosynthesis of various plant secondary metabolites. Among them, 14 DETs and 10 DAMs were enriched in the phenylpropanoid biosynthesis pathway, while 6 DETs and 8 DAMs were enriched in the phenylalanine metabolism pathway in *G. oblongifolia*.

To identify the hub metabolites and transcripts related to xanthone biosynthesis, correlation analysis was performed. First, nine-quadrant diagrams were used to comprehensively evaluate the correlation between metabolites and genes (Fig.S14). Correlation analysis was performed to examine the potential regulatory network between DETs and DAMs. The nine-quadrant graph revealed that several single DAM were regulated by multiple DETs, or a single DET regulated multiple DAMs. In the leaves and bark, there were 5 differentially expressed genes with positive correlation of xanthone biosynthesis, including CoA Hydratase-33,030, CoA Hydratase-50,978, CoA Hydratase-42,503, PT-39,879 and PT-7657. In the fruits and bark, there were 7 differentially expressed genes with positive correlation of xanthone synthesis, including three 4CL transcripts, including 4CL-89,125, 4CL-3808, 4CL-4106, and four PT transcripts, such as PT-7657, PT-90,283, PT-91,156 and PT-86,408. There was no xanthone synthesis-related differential gene in the bark and branches, so it was speculated that the difference of xanthone synthesis between these two tissues was small.

Weighted gene co-expression network analysis (WGCNA) was used to identify highly tissue-associated gene sets. The results indicated that a soft threshold power of 12 was the lowest value with a scale-free topology model fit index of 0.80 and relatively high mean connectivity (Fig.S15). After filtration and selection, 15,404 transcripts were clustered into 17 distinct modules in which the transcripts shared the similar expression patterns (Fig. 6A) and grey module serving to identify unassigned transcripts and containing only one transcript. The black module was the largest, consisting of 3,274 transcripts, followed by the blue4 modules with 1,913 transcripts. In contrast, the antiquewhite2 module contained the fewest transcripts, with only 72 transcripts, while the remaining modules contained between 300 and 1,800 transcripts (Fig. 6B). Among the 17 modules, 7 modules meeting the criteria of $|r|>0.7$ and $P<0.01$ were identified as relevant specific modules for this study, including the brown2, lightcyan, blue4, lightgreen, darkorange, firebrick3, and lightcyan1 modules, which exhibited the highest correlation with the xanthone biosynthesis pathway (Fig. 6C).

The MEbrown2 module (892 transcripts) exhibited a strong positive correlation with the contents of benzophenone (0.7), and the transcripts of BPS-80,339, 4CL-89,125 were clustered into this module. The MElightcyan module (664 transcripts) demonstrated a significant positive correlation with the contents of L-phenylalanine (0.88) and benzaldehyde (0.89), and the transcripts of PT-39,879, bHLH-86,844 and bHLH-101,211 were clustered into this module,. Additionally, the MEblue4 module (1,913 transcripts) displayed a significant positive association with the contents of tripteroside (0.77) and lancerin (0.77), and the transcripts of C4H-58,102 and 4CL-15,993, bHLH-31,167, bHLH-99,805 and bHLH-98,039 were clustered into the MEblue4 module. The MEdarkmagenta module (1,372 transcripts) showed a significant positive correlation with the contents of

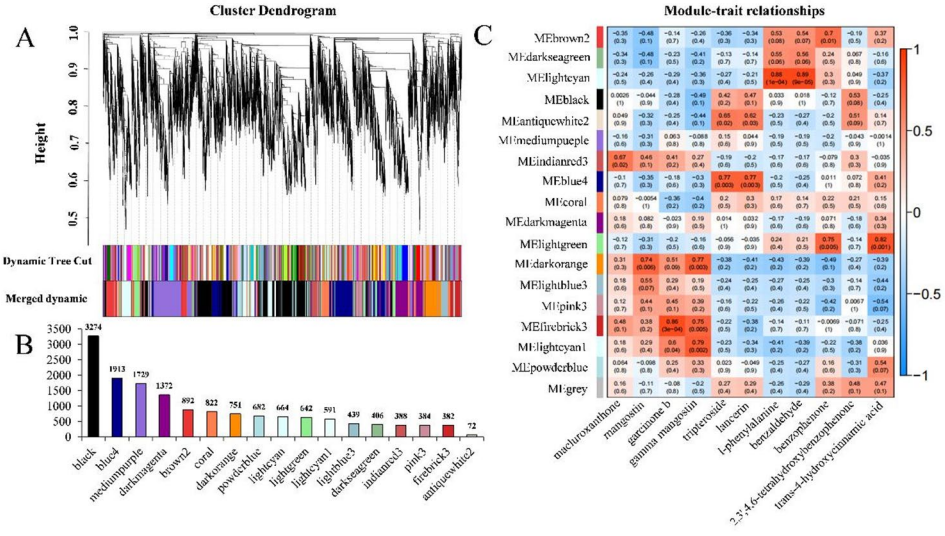


Fig. 6 Weighted gene co-expression network analysis of *G. oblongifolia* **A** The identified gene modules were labeled in different colors **B** The number of genes in each module **C** Module-trait relationship. Correlation coefficient between xanthones, intermediate metabolites in the biosynthetic pathway and module genes presented with a color scale with red and blue representing positive and negative correlations, respectively. A high degree of association between the specific module and trait type was defined:| r |>0.7 and indicated by dark red

benzophenone (0.75) and trans-4-hydroxycinnamic acid (0.82), and the transcripts of BPS-60,300 and XS-4757, bHLH-57,590 and bHLH-94,693 were clustered into the MEdarkmagenta module. Furthermore, the MElightgreen module (642 transcripts) exhibited a significant positive correlation with the contents of mangostin (0.74) and gamma mangostin (0.77), bHLH-4115, bHLH-16,384 and bHLH-59,915 were clustered into the MElightgreen module. The MEpink3 module (384 transcripts) was significantly positively correlated with garcinone b (0.86) and gamma mangostin (0.75). Notably, the MEfirebrick3 module (382 transcripts) was significantly positively correlated with the contents of gamma mangostin (0.79), bHLH-43,456 was clustered into the MElightgreen module. In addition, we identified four modules that were highly related to xanthone biosynthesis. Among our 32 potential genes, 6 synthetic pathway related transcripts, BPS-80,339, 4CL-89,125, PT-39,879, C4H-58,102, BPS-60,300 and XS-4757 belonged to four modules and 11 bHLH transcription factors were clustered into five modules that were highly correlated with xanthone substances, suggesting that they have a certain correlation with the regulation of xanthone, which provided a reference for subsequent gene selection.

Five different types of important genes, they were clustered into different modules because of their differential expression patterns. The study identified key pathways and gene modules involved in the biosynthesis and regulation of various metabolites, which could be further explored for potential applications in biotechnology and plant breeding.

Validation of gene expression by qRT-PCR

To validate the transcriptome sequencing results, we randomly selected nine genes associated with the xanthone biosynthetic pathway and conducted qRT-PCR experiments to assess the reliability of the RNA-Seq findings. The primers used for the qRT-PCR were listed in Table S9, with the actin gene serving as the reference housekeeping gene. We selected actin as the internal reference gene based on its widespread use and well-documented stability in similar plant species and experimental conditions [42, 43]. Meanwhile, the tissue expression profile from our transcriptome data showed that the reference gene actin-33,607 exhibited stable expression across the four tissue sites (Table S10), meeting the requirements for a reference gene. The results revealed that, despite minor discrepancies, the trend observed in the qRT-PCR data was largely in alignment with the RNA-Seq data, thereby reinforcing the credibility of the sequencing results (Fig. 7, Table S11).

Discussion

Structural characteristics and accumulation patterns of xanthone compounds in different tissues of *Garcinia oblongifolia*

Secondary metabolites produced by plants are invaluable sources of medicines, food additives, and industrial products, being widely used for human benefits. The secondary metabolites of *G. oblongifolia* also have various biological activities and pharmacological effects, which have broad prospects in drug development [15, 62, 63]. Xanthone, a pivotal group of plant secondary metabolites, serve as the bioactive compounds in the folk

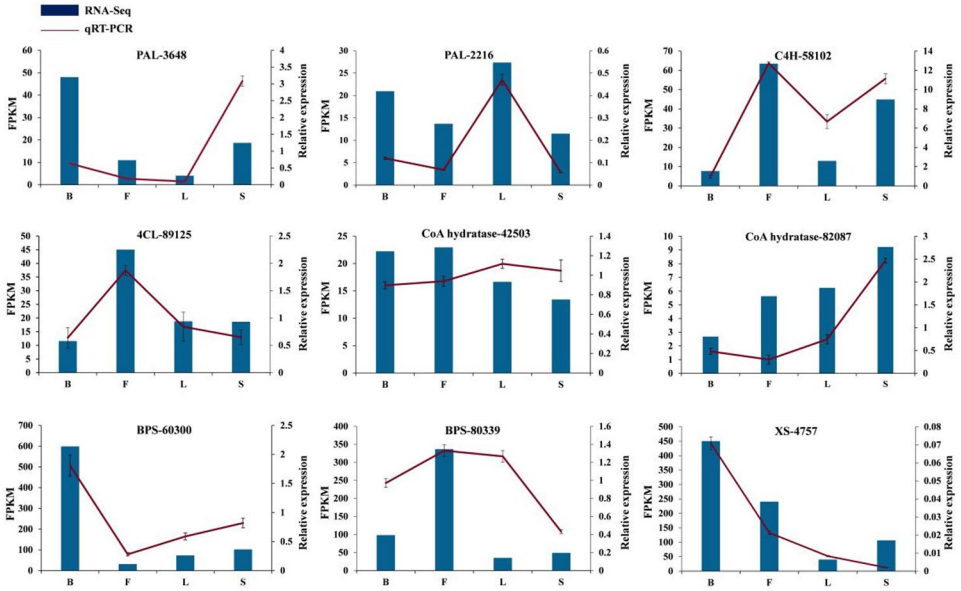


Fig. 7 qRT-PCR validation of nine transcripts related to the xanthone biosynthesis pathway between different tissues. Note: B:bark, F: fruits, L: leaves, S: branches. The left and right ordinates represented the relative expression levels determined by RNA-Seq and qRT-PCR. The relative gene expression of qRT-PCR was evaluated using the $2^{-\Delta\Delta Ct}$ method

medicinal plant *G. oblongifolia*. *Hypericum androsaemum*, *Centaurium erythraea*, and *Garcinia mangostana* are also xanthone-producing plant species, and they have been studied related to xanthone biosynthesis pathways and enzymes [64–66]. In this comprehensive study, an untargeted UPLC-MS/MS metabolomics approach was employed to thoroughly analyze and identify metabolites in four tissues of *G. oblongifolia*. A total of 402 metabolites were detected, with a notable presence of 16 xanthones compounds. The reliability of these findings was underscored by a secondary matching score of 0.8 in metabolite identification. Some xanthones had already been isolated from other species before, for instance, mangiferin was extracted from the roots of *Salacia chinensis* L. and mango leaves [67–69], which was reported to exhibit a multitude of pharmacological properties, including anti-inflammatory [70], antioxidant, antidiabetes [71], cardioprotective [72], and anticancer properties [73]. The compound 1,3,6-trihydroxy-2-(3-methylbut-2-enyl) xanthen-9-one was identified as a potential antitumor drug [74]. Gamma-mangostin (γ-mangostin), derived from *Garcinia mangostana* [75], was predominantly extracted from mangosteen fruits [76]. Mangostin was discovered that accumulated more abundantly in the branches of *G. oblongifolia* [45]. Macluroxanthone was extracted from the leaves of *Garcinia bracteata* [77]. Norathyriol and lancerin isolated from *Tripterospermum lanceolatum* from Gentianaceae had also been documented [78]. Garcinone b was extracted and identified from the fruits of *Garcinia mangostana* in Guttiferae [79]. Dulxanthone f was isolated from the leaves of *G. dulcis* by Kosela et al. in 1999 and mangostinone was the first isolation of a xanthone with a C10 unit from *G. mangostana* [80]. Notably, in our metabolomic study of *G. oblongifolia*, we observed a distinct absence of previously reported xanthone, such as xanthone V2a [81], nigrolineaxanthone T [82], and oblongixanthone I and J [62]. This divergence might be attributed to the geographical variation in the chemical composition and anti-inflammatory activity of the plant [83]. The sample collection area in this study was in Hainan province, in which the content and types of xanthone might be different with those of the materials from Guangdong Province and Guangxi Zhuang Autonomous Region. The plant maturation stage [84] and the different analytical methods [85] also influenced the xanthone identification. Another possible reason for this result could be the absence of the detected compounds in established databases such as the Human Metabolome Database (HMDB) and the Kyoto Encyclopedia of Genes and Genomes (KEGG), which were integral to the annotation process following mass spectrometry analysis. The absence of these compounds in the databases could be due to the novelty of the metabolites. This underscored the importance of continuous updates and expansion of these databases to encompass a broader spectrum of metabolites, particularly those from understudied plant species. By contributing to the knowledge base of *G. oblongifolia* metabolites, our work not only filled gaps in the existing literature but also provided a foundation for future research and potential applications in fields such as pharmacology and natural product chemistry.

Significantly, metabolites accumulated prominently in the bark of *G. oblongifolia*, and differential metabolite analysis revealed that minimal variation between the bark and branches, especially for the 16 xanthones in *G. oblongifolia* which were predominantly concentrated in the branches and bark, rarely distributed in leaves and fruits. We found that there was a correlation between the tissue accumulation of xanthones and their structure. For example, xanthones substituted with isopentenyl groups were mostly distributed in bark and branches, and a few were distributed in fruits (Fig.S5). It is speculated that they were related to the regulation of isopentenyl transferase. Relevant studies had shown that the reaction specificity of isopentenyl enzymes was the basis for the diversity of isopentenyl substances [86]. Pathway analysis further indicated that differential metabolites were primarily enriched in secondary metabolite biosynthesis, phenylpropanoid biosynthesis, and metabolic pathways, consistent with the transcriptome findings.

In addition, based on previous research, we found that the type and position of substituents on xanthone could affect its biological activity. Prior research had indicated that, in addition to the ion-solubilizing characteristics of phenolic groups, the prenylation of xanthones derivatives might modify the hydrophilic-lipophilic balance of the lipid bilayers in the periplasmic membrane of bacteria. Such alterations in membrane properties could increase permeability, potentially leading to cell lysis [87, 88]. This emphasized the significance of geranyl prenylation at the C-8 position and hydroxylation at the C-6 position for optimal antibacterial activity. Conversely, the presence of an additional pyran ring was found to diminish antibacterial effectiveness [88].

Experiments investigating the inhibitory effects of cyclin-dependent kinases (CDKs) had indicated that prenylation at the C-8 position, as opposed to the C-4 position, might enhance the inhibitory activity against CDK4. This enhancement might be attributed to the xanthones' inability to interact with the ATP binding site of CDK4. Conversely, xanthones that possessed an isoprenyl group at the C-8 position, along with hydroxyl groups at the C-1, C-3, and C-6 positions, were identified as the most effective, including α-mangostin (also named as mangostin), β-mangostin, and γ-mangostin. Furthermore, a comparative analysis of α-mangostin, γ-mangostin and garcinone d demonstrated that the hydroxylation of the isoprenyl group could alleviate the negative impact of the methoxy group at C-7 in garcinone d, thereby restoring inhibitory activity against CDK4 and enhancing anticancer efficacy [89]. Studies had also established the significant antioxidative properties of natural xanthones, with 1,3,6-trihydroxy substituents further augmenting antioxidant activity. The prenylation at the C-2 position may not substantially contribute to antioxidant activity, and a prenyl substituent at the C-4 position may diminish antioxidant activity. The antioxidant capacity of isomangiferin and mangiferin provides additional evidence suggesting that the presence of a sugar residue (C-glycosylation) may enhance the antioxidant capacity of xanthones, rather than a prenyl substitution on the C-ring [44, 88]. Analysis revealed that the compound 3, 6, 7, 8 and 10 in the ten prenylated xanthones identified in *G. oblongifolia* contained hydroxyl groups at the C-1, C-3, and C-6 positions (Fig.S10). And among these, the compound 6, 7, and 8 exhibited a prenyl side chain at the C-8 position, so they could show the potential higher antitumor activity. Interestingly, the compound 6, and 8 enriched both in bark and branches, and the compound 3 and 7 also enriched in branches, only the compound 10 enriched in fruits (Fig.S5). These results confirmed the scientific validity of the antitumor activity of the bark of *G. oblongifolia* in the folk, and also the branches could be used as a substitute for the bark as antitumor medicinal use since the branches contained a greater variety and higher content of the active xanthones. This research held promising implications for the medicinal application of *G. oblongifolia*.

Tissue specific expression of enzyme genes related to xanthone biosynthesisand regulation by transcription factors in *Garcinia oblongifolia*

As an important medicinal plant, *G. oblongifolia* has primarily been studied for pharmacological effects and bioactive components. However, there has been little genetic research conducted on *G. oblongifolia*. Hence, further transcriptome and pathway analyses are required to gain a deep insight into the genes involved and their roles in metabolic networking and regulation in valuable secondary metabolites production in *G. oblongifolia*. De novo transcriptome assembly using next-generation sequencing is a fast and economical tool for generating expressed sequence data for non-model plants, particularly those species without available genome sequences [90, 91]. Additionally, single molecule real-time (SMRT) sequencing, currently the most advanced third-generation sequencing technology, has been successfully applied to analyze full-length transcriptomes in multiple plant species, even without a reference genome [92–94].

To generate a more comprehensive transcriptome of *G. oblongifolia*, we combined long-read SMRT sequencing of mixed samples from four tissues with short-read RNA-Seq from bark, fruits, leaves and branches. On the PacBio platform, we obtained a total of 280,945 CCS reads, which yielded 70,401 non-redundant transcripts. After data cleaning and quality assessment of Illumina sequences, a total of 74.11 Gb of clean bases were produced, with each sample yielding more than 5.77 Gb of clean data and 247,443,505 reads. The number of

raw reads was similar or even higher than the previous research data of mangosteen friuts (*Garcinia mangostana* L.) [95, 96]. In our study, the average length of transcripts obtained by combining RNA-Seq and SMRT methods from *G. oblongifolia* was 1,485 bp, which was longer than the average lengths reported in previous studies using only second-generation sequencing platform: 424 bp in *Garcinia indica* [97], 488 bp in *Garcinia mangostana* [98], 784 bp in the seed of *Garcinia mangostana* [99] and 987 bp in *Garcinia xanthochymus* [100], indicating the quality of our data was relatively high compared to the closely related species of *G. oblongifolia* [101, 102]. The results showed that in the absence of a reference genome of *G. oblongifolia*, the integration of RNA-Seq and SMRT analyses provided a more efficient and detailed portrayal of its transcriptome, enabling a comprehensive understanding of the biosynthetic pathway of secondary metabolites.

This study conducted transcriptome analysis on four tissues of *G. oblongifolia*, resulting in the identification of 70,401 non-redundant transcript isoforms, with 53,358 full-length transcripts that received functional annotations from various databases. The analysis revealed 7,818 transcripts showing high homology with *Hevea brasiliensis*. The NR annotation of *G. oblongifolia* transcripts showed high homology with *Hevea brasiliensis* and *Jatropha curcas.* There was limited functional anotation information available for *Garcinia* in public databases, with only 1,046 genes could be retrieved (https://www.ncbi.nlm.nih.gov/gene/?term=Garcinia), even though there were around 40 BioProjects of *Garcinia* in the GenBank database (https://www.ncbi.nlm.nih.gov/bioproject). Additionally, the sequencing studies had predomi nantly concentrated on second-generation sequencing approaches [95, 96]. And the research of *Garcinia* primarily focused on their active compounds and biological activities [8, 44, 62, 82]. Therefore, the NR annotation of *G. oblongifolia* was not matched with the species of *Garcinia*. On the other hand, *Hevea brasiliensis* and *Jatropha curcas* both belong to Euphorbiaceae in Malpighiales, while *G. oblongifolia* belongs to Clusiaceae, which is also in Malpighiales. Other transcriptome studies, such as those on mangosteen, also showed high hit matches with *Jatropha curcas* among the top BLAST matches [103]. The high homology of Clusiaceae and Euphorbiaceaehere is consistent with the APG IV classification system. So, our study conducted the first and most comprehensive transcriptome analysis of *G. oblongifolia*, which is very valuble and could lay the foundation for subsequent genetic research and study of biosynthetic active components in *G. oblongifolia*.

Phylogenetic and motif analysis revealed that *G. oblongifolia* BPS (benzophenone synthase) transcripts-key enzymes in xanthone biosynthesis- clustere closely with those from *G. mangostana* and shared conserved motifs (Motif 1–7), suggesting functional conservation. These enzymes catalyzed benzophenone and xanthone formation, primarily yielding 2,4,6-trihydroxybenzophenone [104]. bHLH TFs are known regulators of plant secondary metabolism, influencing phenylpropanoid, anthocyanin, and lignin biaosynthesis [104–106]. Notably, bHLH TFs might serve as central regulators of BPS expression in xanthone biosynthesis. Tissue-specific expression analysis identified bHLH-14,818, bHLH-16,384, bHLH-21,904, bHLH-13,845, and bHLH-28,903 as highly expressed in fruit or bark tissues-organs critical for plant defense. Among these, bHLH-28,903, which was abundant in bark and branches and implicated in metabolic regulation, emerged as a promising candidate for further functional studies. Xanthones, though understudied, likely contribute to stress responses. Their biosynthesis may intersect with phenylpropanoid and jasmonic acid pathways, which modulate plant defense and flavonoid production [107–110]. Our findings implicate bHLH TFs in regulating xanthone biosynthesis in *G. oblongifolia*, providing a foundation for further research on Garcinia secondary metabolism.

Prediction and regulatory analysis of the xanthone synthesis pathway in *Garcinia oblongifolia*

Integrating metabolome and transcriptome analysis can help comprehensively understand the key functional genes involved in metabolic pathway regulation and their potential molecular mechanisms.Utilizing metabolomics and transcriptomics data, we identified 22 DAMs, including 16 xanthones and 6 intermediate metabolites related to xanthone biosynthesis, and 32 transcripts coding key enzymes in the xanthone biosynthetic pathway across four tissues. By analyzing the transcripts of key enzymes and the intermediate metabolites involved in xanthone synthesis, we established the potential biosynthetic pathway for xanthone. L-phenylalanine, which can be converted from shikimate, is an important substance that links primary and secondary metabolism. It serves as a precursor for many secondary metabolites, such as terpenes, flavonoids, and alkaloids, and it is also the precursor of xanthone [111]. A common response in plants under stress can activate the secondary metabolism pathway [112], leading to the increase of PAL synthesis and the accumulation of corresponding mRNA. Our findings suggested a potential pathway for xanthone biosynthesis in *G. oblongifolia*, through the enzyme PAL, C4H, 4CL, and HCRC to produce benzoyl-CoA, which could be catalyzed by BPS, B3'H and XS to produce xanthone. And another potential pathway began with shikimate, and then was converted to 3-hydroxybenzoic acid and 3-hydroxybenzoyl-CoA, which was catalyzed by BPS, and XS to produce xanthone. Both 1,3,5-triHX and

1,3,7-triHX could be converted by PTs in *G. oblongifolia* into xanthone derivatives which exhibited diverse biological activities, including anti-inflammatory and antitumor effects. Despite the absence of intermediate metabolites in our metabolome data, such as 4-coumaroyl-CoA, 4-hydroxybenzoyl-CoA, benzoyl-CoA, 3-hydroxybenzoyl-CoA, 2,4,6-trihydroxybenzophenone, and 1,3,7-trihydroxyxanthone, we could also speculate the potential synthetic pathways of xanthone, because these compounds could be rapidly catalyzed by the enzymes in the pathway into other substances as soon as they were produced, resulting in no accumulation.

In the biosynthesis pathway, there were two types of enzymes that were not detected in the transcriptome data. One type was ligase, including cinnamate-CoA ligase (CNL), benzoate-CoA ligase (BZL), and 3-hydroxybenzoyl-CoA ligase (3BZL). And the other type was oxidoreductase, including BD, B3'H, and X6H. For the ligases, there were only 6 3BZL genes and none of them were from plants (https://www.ncbi.nlm.nih.gov/gene/?term=3-hydroxybenzoyl-CoA+ligase), and there were 195 BZL genes, with only a few of them were from plants (https://www.ncbi.nlm.nih.gov/gene/?term=benzoate-CoA%20ligase), and there were over 200 CNL genes in the GenBank database (https://www.ncbi.nlm.nih.gov/gene/?term=cinnamate-CoA+ligase) and most of them were from plants. Eventhough there were 70 CoA-synthetase transcripts and 94 CoA-ligase transcripts annotated in our transcriptome data, these three genes were not detected because there was no genes in *G. oblongifolia* or the genes (from *G. oblongifolia*) had little homology with the public database. Then we could screen these transcripts later to discover which of the transcripts could perform the functions of CNL, BZL, and 3BZLin addition to their function of being annotated. For the oxidoreductase, there was no X6H gene and only one B3'H gene in GenBank database which was from *Maclura tricuspidata* in Moraceae belonging to Rosales (https://www.ncbi.nlm.nih.gov/search/all/?term=benzophenone%203%E2%80%99-hydroxylase). There were about 1,100 BD genes in GenBank database (https://www.ncbi.nlm.nih.gov/gene/?term=benzaldehyde+dehydrogenase) and most of them were mitochondrial-like and sourced from DNA sequences. The BD gene in *G. oblongifolia* was not identified, which might also be the case for the CoA ligase genes above. This study provided new insights into the biosynthesis of xanthone in *G. oblongifolia*.

In this study, a significant enrichment of both DETs and DAMs in common KEGG pathways was observed, such as biosynthesis of cofactors, phenylpropanoid biosynthesis, ubiquinone and other terpenoid-quinone biosynthesis, and phenylalanine metabolism after combining metabolome and transcriptome analysis. By comparing the number of DETs and DAMs across different groups, we observed that they shared the similar trends, indicating a certain correlation between differential genes and differential metabolites. Differential expression analysis highlighted a significant number of DETs in the bark compared to other tissues. Specifically, minimal variation was observed between bark and branches, whereas substantial differences were noted between fruits and leaves. This pattern was consistent with the DAMs across tissues.Moreover, among the 16 xanthone compounds we identified, most of them accumulated in the bark and branches. Additionally, the accumulation contents of lichexanthone, norathyriol, cratoxyarborenone in the branches were higher than those in the bark. This result suggested that the branches tissue can be considered as an alternative medicinal part for development and utilization instead of the bark tissue. This notion was supported by studies on *Eucommia ulmoides* Oliv, a plant primarily using bark as its medicinal part, and the research had confirmed that leaves could be a potential alternative medicinal option when *Eucommia ulmoides* Oliv. resources were scarce [113]. The specific DAMs and DETs were involved in xanthone biosynthesis, indicating the existence of multiple metabolic activities of xanthones in *G. oblongifolia*. The results of correlation analysis and WGCNA analysis provided a reference for the screening of genes related to xanthone biosynthesis in *G.oblongifolia*. The findings of this study revealed that the synthesis regulation of xanthone in *G.oblongifolia* involves enzyme genes, offered options for subsequent verification of gene functions.

Among these tissues, some enzyme genes were highly expressed in fruits, such as the enzyme PAL, C4H, and 4CL, which had a positive regulatory relationship with their downstream metabolites (Fig. 4). The PAL enzyme genes detected in this study were mainly highly expressed in fruit parts and were almost not expressed in leaves, which was similar to *Rhododendron fortunei* and other plants [114, 115]. The expression pattern of C4H gene also varied depending on the plant species. C4H enzyme gene in *G. oblongifolia* was highly expressed in branches tissues and the same expression pattern was found in *Nicotiana tabacum* L., in which C4H expressed in the stems, leaves, and flowers, with the highest expression levels in the xylem and phloem of the stems, indicating a close relationship between C4H gene expression and plant lignification process [116, 117]. The BPS enzyme gene was highly expressed in the bark and fruits in this study, and the highest expression level of BPS1 and BPS2 genes in *Garcinia xanthochymus* were observed in similar tissues like sepal and fruit pedicel [100]. The 2,3',4,6-tetrahydroxybenzophenone was the substrate of XS, and it was mainly concentrated in the branches. The XS was highly expressed in the bark and fruits. It was speculated that 2,3',4,6-tetrahydroxybenzophenone

was used more in bark and fruits than other tissues, so the accumulation of it in bark and fruits was lower. As downstream genes in xanthone biosynthesis, most of XS transcripts demonstrated high expression in both bark and fruits so the xanthone could be biosynthesized in the bark and fruits. In the 16 xanthone metabolites, 10 of them contained isoprenyl group. The prenylation reaction of xanthone was catalyzed by prenyltransferases (PTs) [118]. Different prenyltransferases are involved in the biosynthesis and regulation of various substances, including menthol, monoterpene indole alkaloids, and flavonoids [119, 120]. Observations showed that the xanthone compouds with isoprenyl were mainly concentrated in the bark and branches, and the xanthone derivatives withother substituents on xanthone ring concentrated in the fruits or leaves (Fig.S5). The aromatic prenyltransferase of 1,3,6,7-tetrahydroxyxanthone 8-prenyltransferase was screened and it was highly expressed in bark and fruits (Fig. 4). And it had been confirmed that HcPT (*Hypericum calycinum* prenyltransferase) could generate 1,3,6,7-tetrahydroxy-8-prenylxanthone in *Hypericum calycinum* [121]. The difference between the xanthone accumulation pattern and the XS or PT expression pattern indicated that after the synthesis of xanthone in the bark and fruits, xanthone could be transported to branches and leaves for subsequent modification by prenyltransferase, glycosyltransferases, or methyltransferases, and there might be further material migration after modification. The mechanisms underlying the transport and exchange of xanthones between different tissues in *G. oblongifolia* still need further investigation.

Conclusion

In summary, this study employed an integrated transcriptomics and metabolomics approach to investigate the bark, fruits, leaves, and branches of *G. oblongifolia*, a medicinal plant traditionally used for its anti-inflammatory and antitumor properties but with limited existing researches and genomic data. We established a comprehensive metabolic profiling method to elucidate, the tissue-specific distribution of xanthones in *G.oblongifolia*. The identified types and distribution patterns of xanthones across the four tissues provided scientific validation for the traditional use of *G. oblongifolia* bark in antitumor applications. Notably, the xanthone contents in branches was found to be comparable to that in bark, with some branches even exhibiting higher levels of medicinally active xanthones. This suggested that branches could serve as viable alternatives to bark in medicinal application, thereby reducing reliance on bark harvesting and promoting the conservation and sustainable utilization of this plant resources. Additionally, we obtained the first full-length transcriptome data of *G. oblongifolia*, and conducted integrated metabolome and transcriptome analyses to propose a potential biosynthetic pathway for xanthones.Furthermore, bHLH transcription factors highly associated with the regulation of xanthone biosynthesis were identified in *G.oblongifolia*. This study lays a solid foundation for further research on the specific metabolic mechanisms and regulatory strategies of underlying xanthone biosynthesis in *G. oblongifolia*. However, further investigations are required to elucidate the functional roles of identified genes and to evaluate the pharmacological activities of the metabolites. In conclusion, this study reveals the distribution, accumulation and biosynthesis of xanthones in different tissues of *G. oblongifolia*, providing a valuable basis for advancing research in genetic improvement, conservation, sustainable use, biotechnology, and synthetic biology of this medicinal plant.

Abbreviations

SMRT	Single-molecule real-time
PAL	Phenylalanine ammonia lyase
C4H	Cinnamate-4-hydroxylase
4CL	4-coumarate–CoA ligase
BPS	Benzophenone synthase
XS	Xanthone synthase
PT	Prenyltransferase
CNL	Cinnamate: CoA ligase
CHL	Cinnamoyl-CoA hydratase/lyase
BD	Benzaldehyde dehydrogenase
BZL	Benzoate-CoA ligase
3BZL	3-hydroxybenzoate-CoA ligase
B3'H	Benzophenone 3'-hydroxylase
GT	Glycosyltransferases
B	Bark
F	Fruits
L	Leaves
S	Branches
CLES	College of Life and Environmental Sciences
MUC	Minzu University of China
QC	Quality control
UHPLC-MS/MS	Ultra-performance liquid chromatography-tandem mass spectrometry
IDA	Information-dependent acquisition
PCA	Principal Component Analysis
OPLS-DA	Orthogonal Partial Least Squares Discriminant Analysis
VIP	Variable importance in projection
HMDB	Human Metabolome Database
DAMs	Differentially accumulated metabolites
KEGG	Kyoto Encyclopedia of Genes and Genomes
CCS	Circular consensus sequence
FL	Full-length
FLNC	Full-length non-chimeric
ZMW	Zero-mode waveguide
FPKM	Fragments per kilobase of exon per million fragments mapped
FDR	False discovery rate
DETs	Differentially expressed transcripts
NR	Non-redundant protein database
COG	Cluster of Orthologous Groups of proteins
KOG	euKaryotic Orthologous Groups
Pfam	Protein family
eggNOG	Evolutionary Genealogy of genes: Non-supervised Orthologous
TrEMBL	Translation of EMBL
GO	Gene Ontology
KEGG	Kyoto Encyclopedia of Genes and Genomes
WGCNA	Weighted Gene Co-expression Network Analysis

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Supplementary Material 6.

Supplementary Material 7.

Supplementary Material 8.

Supplementary Material 9.

Supplementary Material 10.

Supplementary Material 11.

Supplementary Material 12.

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Authors' contributions
Yue Liu : Supervision, Validation, Writing – review & editing. **Liya Liu** : Investagation, Validation, Writing – review & editing. **Honglei Jin: ** Investagation, Validation, Writing – review & editing. **Bo Liu** : Validation, Writing – review & editing. **Yi Zhang** : Writing – original draft, Methodology, Formal analysis, Data curation. **RuiQi Zhang** : Writing – original draft, Methodology, Formal analysis, Data curation. **Baiyun Wang** : Data curation. **Xinyi Dai** : Data curation. **Xiaoyi Yan** : Data curation. **Xinyue Liu** : Software. **Lu Yang** : Software.

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Data availability
The raw sequence data (full-length transcriptome and RNA-seq) generated in this study are available in the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA1116984. All other data are contained within the article and Supplementary files (accessible via hyperlinks in the manuscript).

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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